| | | | |
|---|---|---|---|
| **Fiscal Year:** | FY 2008 | **Task Last Updated:** | FY 03/17/2008 |
| **PI Name:** | Chang, Hsueh-Chia Ph.D. | | |
| **Project Title:** | Microcirculation Anomalies in Microgravity Blood Flow | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | FLUID PHYSICS | | |
| **Program/Discipline--Element/Subdiscipline:** | FLUID PHYSICS--Complex fluids | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | Hsueh-Chia.Chang.2@nd.edu | **Fax:** | FY |
| **PI Organization Type:** | UNIVERSITY | **Phone:** | 574 631 5697 |
| **Organization Name:** | Notre Dame University | | |
| **PI Address 1:** | 165A Fitzpatrick Engr. | | |
| **PI Address 2:** | | | |
| **PI Web Page:** | | | |
| **City:** | Notre Dame | **State:** | IN |
| **Zip Code:** | 46556 | **Congressional District:** | 3 |
| **Comments:** | | | |
| **Project Type:** | INTERDISCIPLINARY (2004-2007) | **Solicitation / Funding Source:** | 2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities |
| **Start Date:** | 02/04/2002 | **End Date:** | 10/31/2007 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | 4 |
| **No. of PhD Candidates:** | 4 | **No. of Master' Degrees:** | 1 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 1 |
| **No. of Bachelor's Candidates:** | 1 | **Monitoring Center:** | NASA GRC |
| **Contact Monitor:** | Ansari, Rafat | **Contact Phone:** | 216-433-5008 |
| **Contact Email:** | Rafat.R.Ansari@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: Received NCE to 10/31/2007 per PI (11/06) | | |
| **Key Personnel Changes/Previous PI:** | 0 | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | NAG 3 2701 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

| | |
|---|---|
| **Task Description:** | The project is to study anomalies in micro-gravity blood flow in human circulation system and in blood diagnostic kits. We have solved an important blood suspension loading problem into small diagnostic kits and are now working on several chip-scale blood analytical tools. Included are blood coagulation detecion, blood glucose detection, leukemia, sickle cell and blood bacteria detection. The target is to produce simple kits that do not require elaborate equipment that is unavailable in space.<br><br>To develop blood diagnostic kits that can be used in a confined spaceship. The diagnostic kits would be to detect pathogens and anomal physiological symptons duirng prolonged space flights.<br><br>It can produce a robust and miniature bloood diagnostic kit for a variety of blood anomalies and diseases often associated with space flights. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The same blood diagnostic kits could be used as point-of-care and portable diagnostic kits on earth. |
| **Task Progress:** | The project has allowed the PI to launch a ground-breaking study on how blood suspension rheology changes with/without gravity and in different channel dimensions. The results not only allow us to understand microcirculation in space and also in designing blood diagnostics.<br>Specific research achievements include:<br><br>1. Understanding shear-thinning of blood suspensions at high shear rates. This rheological anomaly was attributed to migration of blood cells towards the low shear rate region at the middle of the channel. We are able to collapse literature data on blood suspension viscosity at different flow rates or different pressure drops by a general red blood cell migration theory. This result resolves the classical Fahraeus-Lynquist phenomenon for the first time.<br><br>2. The same phenomenon was shown to prevent loading of blood samples into diagnostic kits with micro-needles less than 100 microns in dimension. The segregated blood cells at the middle of the capillary endows the blood cells with a higher velocity than the liquid and the wetting meniscus. As a result, a dense slug of blood cells builds up behind the meniscus whose high viscosity arrests penetration of the blood suspension into the microneedle.<br><br>3. The development of several AC electrokinetic technologies for identifying blood types, diseased blood cells and blood age. The ion channel activity and the protein concentration on the blood membranes are shown to profoundly affect the dielectrophoretic mobility of the blood cells. Diagnostic kits are then made using this principle to carry out blood diagnostics. The effect of gravitational sedimentation allows for enhanced identification sensitivity with the new design. |
| **Bibliography Type:** | Description: (Last Updated: 02/21/2014) |
| **Articles in Peer-reviewed Journals** | Gordon JE, Gagnon Z, Chang H-C. "Dielectrophoretic discrimination of bovine red blood cell starvation age by buffer selection and membrane cross-linking." Biomicrofluidics 2007 Oct-Dec;1(4):044102. http://dx.doi.org/10.1063/1.2818767 , Oct-2007 |
| **Articles in Peer-reviewed Journals** | Chang H-C. "Nanobead electrokinetics: The enabling microfluidic platform for rapid multi-target pathogen detection." AIChE Journal 2007 Oct;53(10):2486-92. http://dx.doi.org/10.1002/aic.11286 , Oct-2007 |
| **Articles in Peer-reviewed Journals** | Cheng I-F,Chang H-C, Hou D, Chang H-C. "An integrated dielectrophoretic chip for continuous bioparticle filtering, focusing, trapping and detecting." Biomicrofluidics, 2007 Apr-Jun;1(2):021503. http://dx.doi.org/10.1063/1.2723669 , Apr-2007 |
| **Articles in Peer-reviewed Journals** | Hou D, Chang H-C. "Electrokinetic particle aggregation patterns in microvortices due to particle-field interaction." Physics of Fluids 2006 Jul;18(6):071702. http://dx.doi.org/10.1063/1.2221348 , Jul-2006 |
| **Articles in Peer-reviewed Journals** | Sengupta S, Battigelli DA, Chang HC. "A micro-scale multi-frequency reactance measurement technique to detect bacterial growth at low bio-particle concentrations." Lab Chip. 2006 May;6(5):682-92. PMID: 16652185 , May-2006 |
| **Articles in Peer-reviewed Journals** | Minerick AR, Zhou R, Takhistov P, Chang HC. "Manipulation and characterization of red blood cells with alternating current fields in microdevices." Electrophoresis. 2003 Nov;24(21):3703-17. PMID: 14613196 , Nov-2003 |
| **Articles in Peer-reviewed Journals** | Minerick AR, Chang HC, Hoagland TM, Olson KR. "Dynamic synchronization analysis of venous pressure-driven cardiac output in rainbow trout." Am J Physiol Regul Integr Comp Physiol. 2003 Oct;285(4):R889-96. PMID: 12805092 , Oct-2003 |